

Calcolo energia - area regole:

1

$$x(t) = (1+i) e^{-\frac{2|t|}{T} + i2\pi \frac{|t|}{T}}$$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

Per calcolare il modulo di $(a+ib) e^{d+ib}$: $\sqrt{a^2+b^2} e^d$

$$|x(t)| = \sqrt{1^2+1^2} e^{-\frac{2|t|}{T}} = \sqrt{2} e^{-\frac{2|t|}{T}}$$

$$|x(t)|^2 = 2 e^{-\frac{4|t|}{T}}$$

$$E_x = \int_{-\infty}^{+\infty} 2 e^{-\frac{4|t|}{T}} dt$$

$x(t) = x(-t) \rightarrow$ segnale pari

$$E_x = 2 \int_0^{+\infty} 2 e^{-\frac{4t}{T}} dt = 4 \int_0^{+\infty} e^{-\frac{4t}{T}} dt = -\frac{4T}{1} \int_0^{+\infty} e^{-\frac{4t}{T}} = -T e^{-\frac{4t}{T}} \Big|_0^{+\infty} =$$

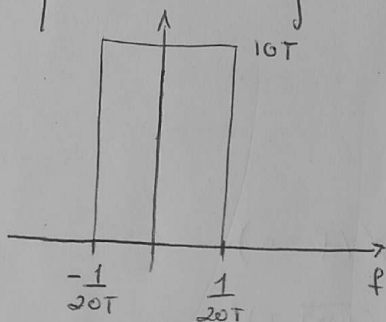
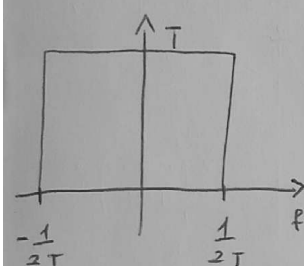
$$= T(0-1) = T$$

$$s(t) = V_0 \operatorname{sinc} \frac{t}{T} \operatorname{sinc} \frac{0.1t}{T} = V_0 \operatorname{sinc} \frac{t}{T} \operatorname{sinc} \frac{t}{10T}$$

$$E_s = \int_{-\infty}^{+\infty} |s(t)|^2 dt = \int_{-\infty}^{+\infty} |x(f)|^2 df$$

↑
parassero

$$X(f) = V_0 \cdot \left\{ T \operatorname{rect} fT \right\} * \left\{ 10T \operatorname{rect} 10fT \right\}$$

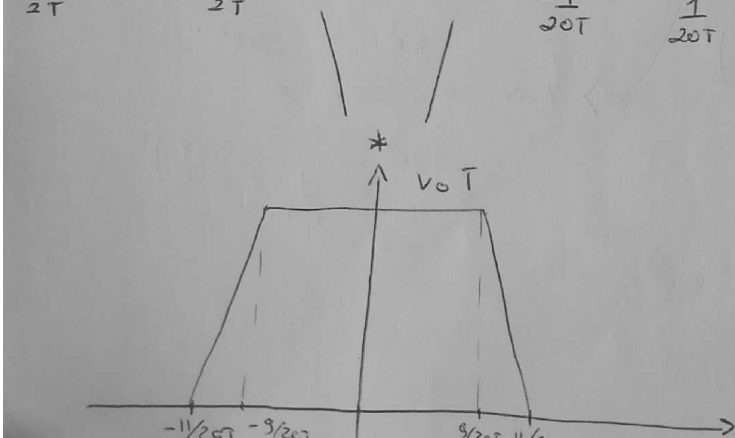


$T_1 > T_2$

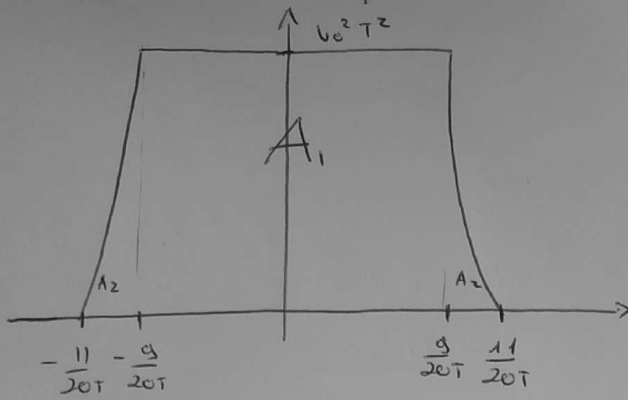
$$B = 2 \cdot \left(\frac{1}{2T} + \frac{1}{20T} \right) = 2 \cdot \left(\frac{10+1}{20T} \right) = 2 \cdot \left(\frac{11}{20T} \right)$$

$$b = 2 \cdot \left(\frac{1}{2T} - \frac{1}{20T} \right) = 2 \cdot \left(\frac{9}{20T} \right)$$

$$B = A_1 A_2 T_2 = T \cdot 10T \cdot \frac{1}{10T} = T$$



Cominciamo con il quadrato di $x(f)$: $\rightarrow$ Gli intervalli non cambiano
 $\rightarrow$ i lati obliqui diventano rami di parabola
 $\rightarrow$ le ampiezze di ventano il quadrato



Calcoliamo l'area del segnale sommando i contributi:

$$A = A_1 + A_2$$

$$= \left(2 \cdot \frac{9}{20T}\right) \cdot V_0^2 T^2 + 2 \cdot \left(\frac{11}{20T} - \frac{9}{20T}\right) \cdot \frac{V_0^2 T^2}{3}$$

$$= \frac{9}{10} V_0^2 T + \frac{2}{10} \frac{4}{20} \frac{V_0^2 T}{3} = \frac{9}{10} V_0^2 T + \frac{2}{30} V_0^2 T = \frac{27+2}{30} V_0^2 T = \frac{29}{30} V_0^2 T$$

$$x(t) = (1+2i) e^{-2\frac{(t-3)^2}{T}} + \frac{i|t|}{T}$$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

$$|x(t)| = \sqrt{1^2 + 2^2} e^{-2\frac{(t-3)^2}{T}}$$

$$|x(t)|^2 = 5 e^{-4\frac{(t-3)^2}{T}}$$

$$E_x = \int_{-\infty}^{+\infty} 5 e^{-4\frac{(t-3)^2}{T}} dt = \int_{-\infty}^{+\infty} 5 e^{-4\frac{t^2}{T}} dt$$

invarianza rispetto alle traslazioni

$$y(x) = 5 e^{-\frac{4x^2}{T}}$$

$$\text{area } y(x) = \gamma(f=0)$$

$$5 e^{-\frac{\pi \left(\frac{2x}{\sqrt{\pi T}}\right)^2}{2}} \longleftrightarrow \frac{5\sqrt{\pi T}}{2} e^{-\frac{\pi \left(\frac{\sqrt{\pi T}}{2} f\right)^2}{2}}$$

$$\Rightarrow \text{NB } e^{-\frac{\pi t^2}{2}} \Leftrightarrow e^{-\frac{\pi f^2}{2}}$$

$$e^{-\frac{\pi (at)^2}{2}} = \frac{1}{|a|} e^{-\frac{\pi \left(\frac{t}{a}\right)^2}{2}}$$

$$\gamma(0) = E_x = \frac{5\sqrt{\pi T}}{2}$$

$$x(t) = e^{-\pi t^2} \quad y(t) = \text{sinc}\left(\frac{t}{T}\right)$$

$$z(t) = x * y(t)$$

$$\bar{E}_z = \int_{-\infty}^{+\infty} |z(t)|^2 dt = \int_{-\infty}^{+\infty} |z(f)|^2 df$$

Parseval

$$z(f) = x(f) y(f)$$

$$x(f) = e^{-\pi f^2} \quad y(f) = T \text{rect}(fT)$$

$$\bar{E}_z = \int_{-\infty}^{+\infty} T^2 e^{-2\pi f^2} \text{rect}(fT) df = \int_{-\frac{1}{2T}}^{\frac{1}{2T}} T^2 e^{-2\pi f^2} df$$

Questo si tratta di un integrale di una densità di probabilità gaussiana:

$$\int_a^b \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{f^2}{2\sigma^2}} df = \Phi(b) - \Phi(a)$$

$$\frac{f^2}{2\sigma^2} = +2\pi f^2 \Rightarrow \sigma^2 = \frac{1}{4\pi}$$

Faccendo un cambio di variabile: $\sigma^2 = \frac{1}{4\pi}$

$$\bar{E}_z = \frac{T^2}{\sqrt{2}} \int_{-\frac{1}{2T}}^{\frac{1}{2T}} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{f^2}{2\sigma^2}} df = \frac{T^2}{\sqrt{2}} \left[\Phi\left[\frac{1}{2T}\right] - \Phi\left[-\frac{1}{2T}\right] \right]$$

$$x(t) = (1+i) e^{-2t-3+it} 1(t)$$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

$$|x(t)| = \sqrt{1+1} e^{-2t-3} 1(t) = \sqrt{2} e^{-2t-3} 1(t) \quad |x(t)|^2 = 2 e^{-4t-6} 1(t)$$

$$E_x = \int_{-\infty}^{+\infty} 2 e^{-4t-6} 1(t) dt = \int_{-\infty}^{+\infty} 2 e^{-4t-6} 1(t) dt = e^{-6} \int_0^{+\infty} 2 e^{-4t} dt = -\frac{e^{-6}}{2} e^{-4t} \Big|_0^{+\infty} =$$

$$= -\frac{e^{-6}}{2} (0 - 1) = \frac{e^{-6}}{2}$$

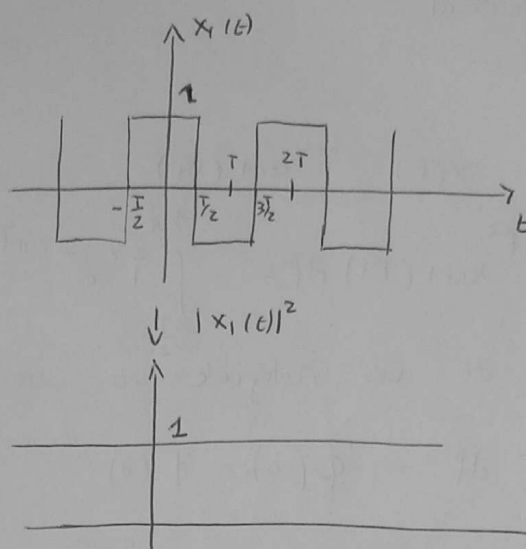
$$X(t) = \sum_{k=-\infty}^{+\infty} (-1)^k e^{-t^2} \text{rect}\left(\frac{t-kT}{T}\right) 1(t)$$

$$= e^{-t^2} 1(t) \sum_{k=-\infty}^{+\infty} \text{rect}\left(\frac{t-kT}{T}\right)$$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt$$

$$x(t) = e^{-t^2} 1(t) X_1(t)$$

$$\int_{-\infty}^{+\infty} |X_1(t)|^2 dt = 1 \quad \text{quindi}$$



$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt =$$

$$= \int_{-\infty}^{+\infty} |e^{-t^2} 1(t) X_1(t)|^2 dt = \int_{-\infty}^{+\infty} |e^{-t^2} 1(t)|^2 dt = \int_0^{+\infty} e^{-2t^2} dt = \frac{1}{2} \int_{-\infty}^{+\infty} e^{-2t^2} dt$$

$$e^{-2t^2} = e^{-\pi \left(\frac{\sqrt{2}}{\sqrt{\pi}} t\right)^2} \longleftrightarrow \frac{\sqrt{\pi}}{\sqrt{2}} e^{-\pi \left(\frac{\sqrt{\pi}}{\sqrt{2}} f\right)^2}$$

L'area dell'integrale: $\frac{1}{2} \int_{-\infty}^{+\infty} e^{-2t^2} dt$ è pari al valore assunto dalla trasformata di Fourier di e^{-2t^2} nell'origine, quindi:

$$E_x = \frac{1}{2} \int_{-\infty}^{+\infty} e^{-2t^2} dt = \frac{1}{2} \frac{\sqrt{\pi}}{\sqrt{2}}$$

$$x(t) = -(t-1) e^{-\pi(t-1)^2}$$

$$A_x = \int_{-\infty}^{+\infty} x(t) dt = \int_{-\infty}^{+\infty} -(t-1) e^{-\pi(t-1)^2} dt = \int_{-\infty}^{+\infty} -t e^{-\pi t^2} dt = 0$$

invarianza rispetto alle traslazioni

$$x(t) = -x(-t)$$

$$-t e^{-\pi t^2} = -(-t e^{-\pi t^2})$$

il segnale è dispari!

NB: l'area di un segnale dispari è zero

Calcolo dell'energia di: $X(f) = \frac{T(1-0,1^2)}{|1-0,1e^{-j2\pi fT}|^2}$

3

Sapendo che $X(f) = X_1(f) + X_1(-f) - T$ con $X_1(f) = \frac{T}{1-0,1e^{-j2\pi fT}}$

Possiamo scrivere

$E_{X(kT)} = T \sum_{k=-\infty}^{+\infty} |X(kT)|^2$ il problema è determinare $X(kT)$ a partire da $X(f)$.

$$X(f) = \frac{T}{1-0,1e^{-j2\pi fT}} + \frac{T}{1-0,1e^{j2\pi fT}} - T$$

$$= T \sum_{k=0}^{+\infty} 0,1^k e^{-j2\pi f k T} + T \sum_{k=0}^{+\infty} 0,1^k e^{j2\pi f k T} - T$$

$$= T \sum_{k=-\infty}^{+\infty} 0,1^{|k|} e^{-j2\pi f k T} \quad \text{da cui} \Rightarrow X(kT) = (0,1)^{|k|}$$

$$E_X = T \sum_{k=-\infty}^{+\infty} |X(kT)|^2 = T \sum_{k=-\infty}^{+\infty} 0,1^{2|k|} = T \sum_{k=0}^{+\infty} 0,1^{2k} + T \sum_{k=0}^{+\infty} 0,1^{2k} - T =$$

$$= \frac{T}{1-0,1^2} + \frac{T}{1-0,1^2} - T = \frac{2T}{1-0,01} - T$$

Calcolo dell'energia di $x(t) = 5te^{-\frac{t^2}{4} + 4j}$

$$E_X = \int_{-\infty}^{+\infty} |x(t)|^2 dt \quad |x(t)| = 5te^{-\frac{t^2}{4}} \quad |x(t)|^2 = 25t^2 e^{-\frac{t^2}{2}}$$

$$= 25t^2 e^{-\frac{1}{2}t^2}$$

$$E_X = \int_{-\infty}^{+\infty} 25t \cdot te^{-\frac{t^2}{2}} dt = 2 \int_0^{+\infty} -25te^{-\frac{t^2}{2}} dt + 2 \int_0^{+\infty} 25te^{-\frac{t^2}{2}} dt =$$

↑
integrando per parti

$$\begin{aligned} f &= 25t & f' &= -25 \\ g &= -e^{-\frac{t^2}{2}} & g' &= te^{-\frac{t^2}{2}} \end{aligned}$$

$$= 0 + 25 \int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt$$

→ utilizziamo la trasformata di Fourier dell'impulso gaussiano →

$$E_x = \int_{-\infty}^{+\infty} 25 e^{-\frac{t^2}{2}} dt$$

$$x(x) = e^{-\frac{t^2}{2}} = e^{-\pi \left(\frac{t}{\sqrt{2\pi}}\right)^2} \quad \longleftrightarrow \quad S(f) = \sqrt{2\pi} e^{-\pi \left(\frac{\sqrt{2\pi}}{2\sqrt{2}} f\right)^2}$$

$$E_x = \int_{-\infty}^{+\infty} 25 e^{-\frac{t^2}{2}} dt = 25 S(f=0) = 25 \sqrt{2\pi}$$

Calcolo dell'energia di $x(t) = 2j e^{-4(t-1)^2 + 2j}$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt \quad |x(t)| = \sqrt{2} e^{-4(t-1)^2} = 2 e^{-4(t-1)^2}$$

$$|x(t)|^2 = 4 e^{-8(t-1)^2}$$

$$E_x = \int_{-\infty}^{+\infty} 4 e^{-8(t-1)^2} dt = \int_{-\infty}^{+\infty} 4 e^{-8t^2} dt = 4 \int_{-\infty}^{+\infty} e^{-8t^2} dt \rightarrow \text{utilizziamo la trasformata di Fourier dell'impulso gaussiano:}$$

invarianza rispetto alle traslazioni

$$x(t) = e^{-8t^2} = e^{-\pi \left(\frac{\sqrt{8}}{\sqrt{\pi}} t\right)^2} \quad \longleftrightarrow \quad \frac{\sqrt{\pi}}{\sqrt{8}} e^{-\pi \left(\frac{\sqrt{\pi}}{\sqrt{8}} f\right)^2} = \frac{\sqrt{\pi}}{2\sqrt{2}} e^{-\pi \left(\frac{\sqrt{\pi}}{2\sqrt{2}} f\right)^2}$$

$$E_x = 4 \int_{-\infty}^{+\infty} e^{-8t^2} dt = 2 \cdot \frac{\sqrt{\pi}}{2\sqrt{2}} e^{-\pi \left(\frac{\sqrt{\pi}}{2\sqrt{2}} f\right)^2} \bigg|_{f=0} = \frac{2\sqrt{\pi}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \sqrt{2\pi}$$

Calcolo dell'energia di $x(t) = (1+i) e^{-t^2+2t}$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt \quad |x(t)| = \sqrt{2} e^{-t^2+2t} \quad |x(t)|^2 = 2 e^{-2t^2+4t}$$

La faccio diventare un quadrato:

$$E_x = \int_{-\infty}^{+\infty} 2 e^{-2(t-1)^2} dt = e^2 \int_{-\infty}^{+\infty} e^{-2t^2} dt \rightarrow \text{utilizzo la transf. di Fourier dell'impulso gaussiano}$$

$$-2t^2+4t = -2(t^2-2t+1-1) = -2(t^2-2t+1) + 2 = -2(t-1)^2 + 2$$

invarianza rispetto alle traslazioni

$$x(t) = e^{-2t^2} = e^{-\pi \left(\frac{\sqrt{2}}{\sqrt{\pi}} t\right)^2} \quad \longleftrightarrow \quad \frac{\sqrt{\pi}}{\sqrt{2}} e^{-\pi \left(\frac{\sqrt{\pi}}{\sqrt{2}} f\right)^2}$$

$$E_x = e^2 \frac{\sqrt{\pi}}{\sqrt{2}} e^{-\pi \left(\frac{\sqrt{\pi}}{\sqrt{2}} f\right)^2} \bigg|_{f=0} = e^2 \frac{\sqrt{\pi}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{e^2 \sqrt{2\pi}}{2}$$

Calcolo della potenza di $s(t) = V_0 \cos(2\pi f_0 t + \frac{\pi}{5})$ (4)

$$P_s = \lim_{T \rightarrow +\infty} \frac{1}{2T} \int_{-T}^T |s(t)|^2 dt$$

$$= \lim_{T \rightarrow +\infty} \frac{1}{2T} \int_{-T}^T V_0^2 \cos^2(2\pi f_0 t + \frac{\pi}{5}) dt \rightarrow \text{Ho una funzione periodica di periodo } T_p = \frac{1}{f_0}, \text{ integro in quel periodo}$$

$$= \lim_{K \rightarrow +\infty} \frac{1}{2KT_p} \int_{-KT_p}^{KT_p} V_0^2 \cos^2(2\pi f_0 t + \frac{\pi}{5}) dt$$

NB

$$\left[\begin{aligned} \cos(a) \cos(b) &= \frac{1}{2} [\cos(a+b) \cos(a-b)] \\ \cos(2\pi f_0 t + \frac{\pi}{5}) \cos(2\pi f_0 t + \frac{\pi}{5}) &= \frac{1}{2} [\cos(4\pi f_0 t + \frac{2\pi}{5}) + \cos(0)] \\ &= \frac{1}{2} \cos(4\pi f_0 t + \frac{2\pi}{5}) + \frac{1}{2} \end{aligned} \right]$$

$$P_s = \lim_{K \rightarrow +\infty} \frac{1}{2KT_p} \int_0^{T_p} \frac{V_0^2}{2} + V_0^2 (4\pi f_0 t + \frac{2\pi}{5}) dt = \frac{V_0^2}{2}$$

↑
integro su un periodo

Calcolo dell'area di $x(t) = \text{sinc}^4(2t-1)$

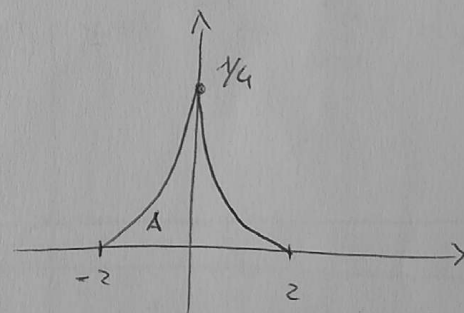
$$A_x = \int_{-\infty}^{+\infty} x(t) dt = \int_{-\infty}^{+\infty} \text{sinc}^4(2t-1) dt = \int_{-\infty}^{+\infty} \text{sinc}^4(2t) dt =$$

$$= \int_{-\infty}^{+\infty} \underbrace{\text{sinc}^2(2t)}_{x_1} \underbrace{\text{sinc}^2(2t)}_{x_2} dt = \int_{-\infty}^{+\infty} \frac{1}{2} \text{tri}\left(\frac{f}{5}\right) \cdot \frac{1}{2} \text{tri}\left(\frac{f}{5}\right) df$$

↑
invarianza + traslazione

Parseval: $\xrightarrow{\text{Parseval}}$

$$\int_{-\infty}^{+\infty} x_1(t) x_2^*(t) dt = \int_{-\infty}^{+\infty} x_1(f) x_2^*(f) df$$



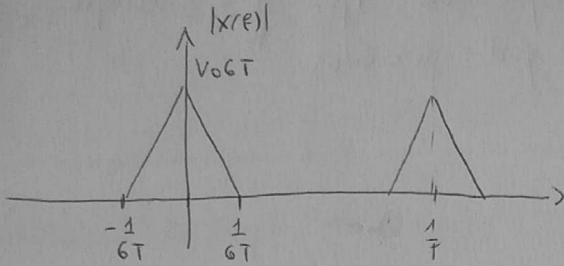
$$A = \frac{b \cdot h}{3} = (2+2) \cdot \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{3}$$

Calcolo dell'energia di $x(kT) = V_0 \operatorname{sinc}^2\left(\frac{T-kT}{6T}\right)$

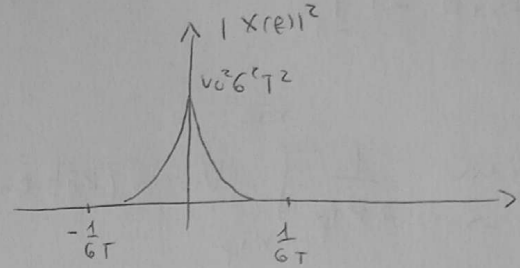
$$E_x = \text{area} \left\{ V_0^2 \operatorname{sinc}^4\left(\frac{kT}{6T}\right) \right\} = \int_{R/2(1/T)} df |X(f)|^2$$

Parseval

$$X(f) = \operatorname{rect}_{\frac{1}{T}} \left\{ V_0 6T + j\pi(6Tf) \right\}$$



→



$$E_x = \frac{2}{6T} \cdot \frac{V_0^2 6^2 T^2}{3} = 4V_0^2 T$$

Calcolare l'area di $x(t) = t^2 e^{-\pi t^2}$

$$A_x = \int_{-\infty}^{+\infty} x(t) dt = \int_{-\infty}^{+\infty} t^2 e^{-\pi t^2} dt = \int_{-\infty}^{+\infty} t \cdot t e^{-\pi t^2} dt =$$

$$= \left. -\frac{2t}{2\pi} e^{-\pi t^2} \right|_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} \frac{1}{2\pi} e^{-\pi t^2} dt = \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-\pi t^2} dt$$

$$X'(f) = e^{-\pi f^2}$$

$$A_x = \left. \frac{1}{2\pi} e^{-\pi f^2} \right|_{f=0} = \frac{1}{2\pi}$$

f	f'
t	1
g	g'
$-\frac{1}{2\pi} e^{-\pi t^2}$	$t e^{-\pi t^2}$

Calcolare l'energia di $x(kT) = 0,5^{1k-11} 1_0(kT-5T)$

$$E_x = T \sum_{k=0}^{+\infty} 0,5^{2(1k-11)} = T \sum_{k=0}^{+\infty} 0,5^{2k} 0,5^2 = 0,5^2 T \frac{1}{1-0,5^2} = \frac{0,5^2 T}{1-0,5^2}$$

Calcolo area ed energia di $x(kT) = (-1)^k e^{-2|k|T}$

(5)

Esso è il segnale discreto:

$$A_x = T \sum_{k=-\infty}^{+\infty} x(kT)$$

$$= T \sum_{k=-\infty}^{+\infty} (-1)^k e^{-2|k|T} = T \left(\sum_{k=0}^{+\infty} (-1)^k e^{-2kT} + \sum_{k=0}^{+\infty} (-1)^{-k} e^{-2kT} - 1 \right)$$

$$= T \left(\frac{1}{1 + e^{-2T}} + \frac{1}{1 + e^{-2T}} - 1 \right) = \frac{2T}{1 + e^{-2T}} - T$$

$$E_x = T \sum_{k=-\infty}^{+\infty} |x(kT)|^2 = T \sum_{k=-\infty}^{+\infty} (-1)^{2k} e^{-4|k|T} =$$

$$= T \left(\sum_{k=0}^{+\infty} (-1)^{2k} e^{-4kT} + \sum_{k=0}^{+\infty} (-1)^{-2k} e^{-4kT} - 1 \right) =$$

$$= T \left(\frac{1}{1 - e^{-4T}} + \frac{1}{1 - e^{-4T}} - 1 \right) = \frac{2T}{1 - e^{-4T}} - T$$

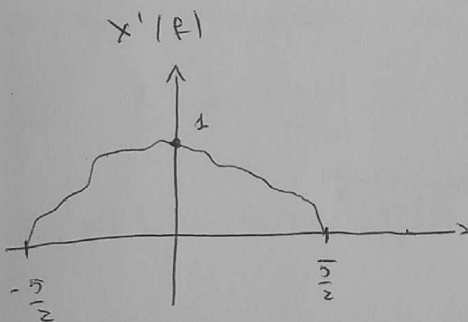
Calcolo minima freq. di campionamento

Calcolo della minima frequenza di campionamento di:

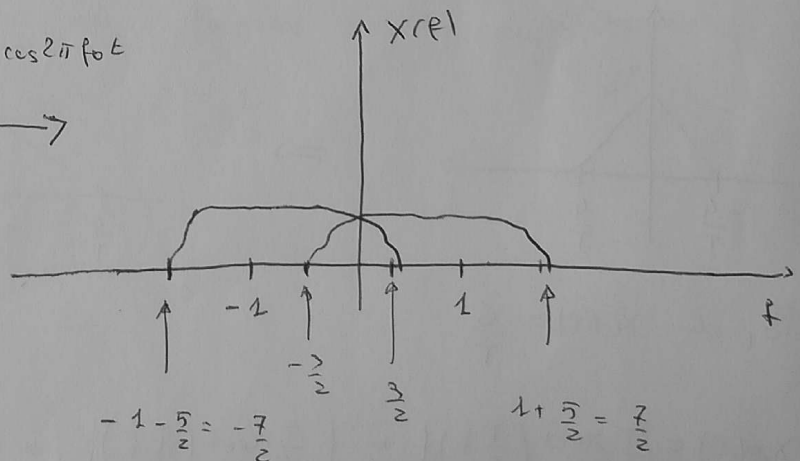
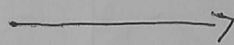
$$x(t) = \underbrace{\text{sinc}^5(t)}_{x'(t)} \cos 2\pi t \quad t \in \mathbb{R}$$

Per il teo del campionamento $F_c \geq 2B$

$x'(t)$ in frequenza non è altro che la convoluzione di 5 rect con $B = \frac{5}{2}$. $\rightarrow f_0 = 1$



effetto del $\cos 2\pi f_0 t$



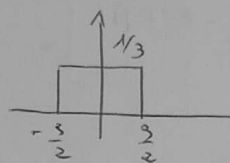
Il segnale risultante ha $B = \frac{7}{2}$

Quindi $F_c \geq 2B \Rightarrow F_c \geq 7$ campioni/s

Calcolo frequenza di campionamento di $x(t) = \sin^3(2-3t)$

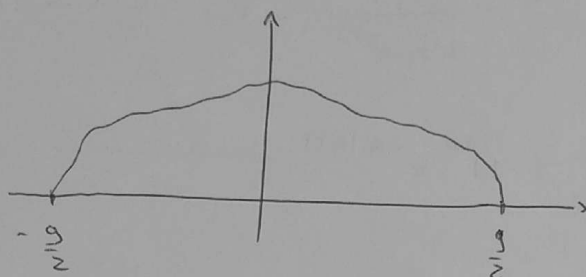
Non considero l'effetto della traslazione

in frequenza il mio segnale si riduce ad una convoluzione di 3 segnali di tipo $\frac{1}{3} \text{rect}\left(\frac{f}{3}\right)$



La Banda del segnale sarà quindi:

$$\frac{3}{2} \cdot 3 = \frac{9}{2} = B$$



$$F_c \gg 2B \Rightarrow F_c \gg 9 \text{ campioni/sec}$$

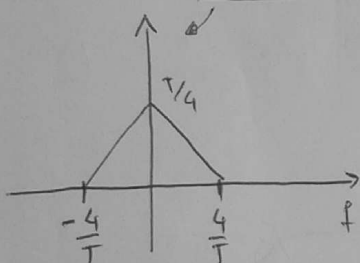
Calcolo frequenza di campionamento $x(t) = \sin^2\left(\frac{4(t+0,2T)}{T}\right) \cos\left(\frac{4\pi t}{T}\right) + \sin^2\left(\frac{4t}{T}\right)$

$$x_1(t) = \sin^2\left(\frac{4}{T}(t+0,2T)\right) \cos\left(\frac{4\pi t}{T}\right)$$

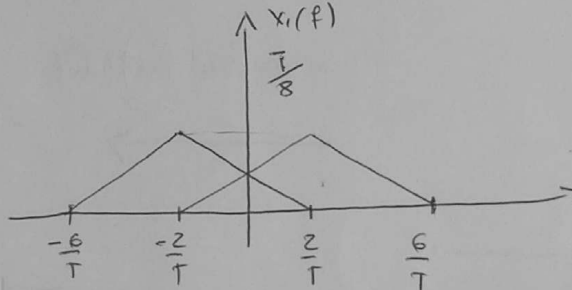
$$\cos\left(\frac{4\pi t}{T}\right) \rightarrow \cos(2\pi f_0 t) \quad f_0 = \frac{2}{T}$$

$\sin^2\left(\frac{4t}{T} + 0,2\right) \rightarrow$ non considero l'effetto della traslazione: $\sin^2\left(\frac{4t}{T}\right)$

$$X_1(f) = \underbrace{\frac{T}{4} \text{tri}\left(\frac{T}{4}f\right)}_{\text{triangular}} * \left(\delta_R\left(f - \frac{2}{T}\right) + \delta_R\left(f + \frac{2}{T}\right)\right) \cdot \frac{1}{2}$$



$\Rightarrow$



$$B_1 \text{ di } X_1(f) = \frac{6}{T}$$

$$X_2(f) = \left\{ \frac{T}{4} \text{rect}\left(\frac{T}{4}f\right) \right\} * \left\{ \frac{T}{4} \text{rect}\left(\frac{T}{4}f\right) \right\} * \left\{ \frac{T}{4} \text{rect}\left(\frac{T}{4}f\right) \right\}$$

$$\text{convoluzione di 3 rect con } B = \frac{2}{T} \Rightarrow B_2 = 3 \cdot \frac{2}{T} = \frac{6}{T} \rightarrow$$

$$B_1 = B_2 = \frac{6}{T}$$

$$T_c \geq 2B \Rightarrow T_c \geq \frac{12}{T} \text{ campioni/s}$$

Calcolo Trasformate di Fourier

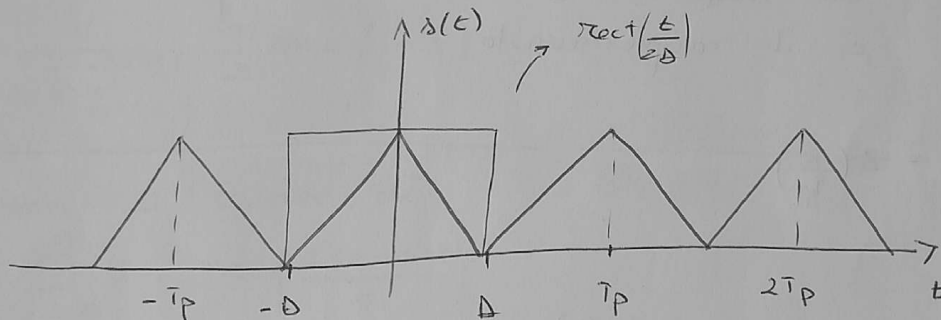
$$s(t) = \sum_{k=-\infty}^{+\infty} \left(1 - \frac{|t - kT_p|}{B}\right) \text{rect}\left(\frac{-t + kT_p}{2B}\right) \quad t \in \mathbb{R}/\mathbb{Z}(T_p)$$

$$B \leq \frac{T_p}{2}$$

$$I = \mathbb{R}/\mathbb{Z}(T_p) \rightarrow \hat{I} = \mathbb{Z}\left(\frac{1}{T_p}\right)$$

$$s(t) = \sum_{k=-\infty}^{+\infty} \left(1 - \frac{|t - kT_p|}{B}\right) \text{rect}\left(\frac{t - kT_p}{2B}\right) \Rightarrow \text{il rect \u00e8 invariante rispetto al ribaltamento degli assi}$$

Per capire il tipo di segnale cui ci dobbiamo a che fare, proviamo a rappresentarlo graficamente ottenendo:



Si tratta di una periodizzazione di $\text{tri}\left(\frac{t}{B}\right)$.

Essendo il campionamento la trasformazione duale della periodizzazione, otteniamo la trasformata di Fourier del campionamento

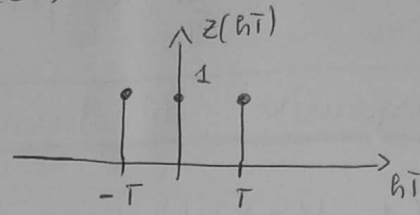
$$\text{di } \mathcal{F}\left[\text{tri}\left(\frac{t}{B}\right)\right] = B \text{sinc}^2(fB) \quad f = \frac{1}{T_p}$$

$$X(kf) = B \text{sinc}^2(kfB) = B \text{sinc}^2\left(\frac{k}{T_p}B\right)$$

Trasformata di Fourier di:

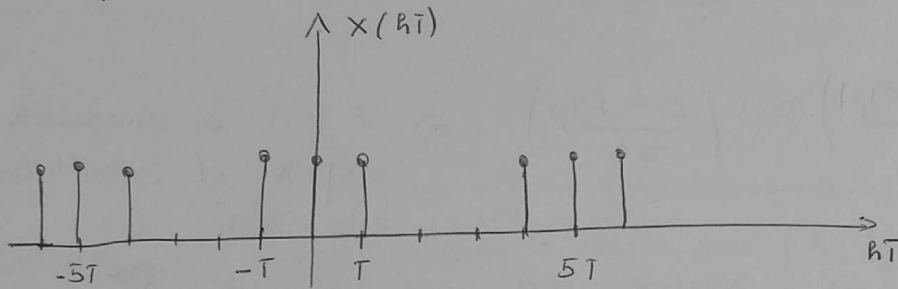
$$X(hT) = \sum_{k=-\infty}^{+\infty} z(hT - kT) \quad hT \in \mathbb{Z}T / \mathbb{Z}(5T)$$

$$z(hT) = \begin{cases} 1 & h = -1, 0, 1 \\ 0 & \text{altrove} \end{cases}$$



$$I = \mathbb{Z}T / \mathbb{Z}(5T) \xrightarrow{f} \hat{I} = \mathbb{Z}(\frac{1}{5T}) / \mathbb{Z}(\frac{1}{T})$$

$X(hT)$ è la periodizzazione di $z(hT)$:



$$z(hT) \xrightarrow{T_p = 5T} X(hT) = z(hT - kT_p)$$

La trasformazione duale della periodizzazione è il campionamento:

$$z(f) \xrightarrow{T_p = 5T} X(f) = z\left(\frac{k}{T_p}\right)$$

$$z(f) = \mathcal{F}[z(hT)]$$

$$z(hT) = \text{rect}\left(\frac{h}{3T}\right)$$

$$z(f) = 3T \text{sinc}(3Tf)$$

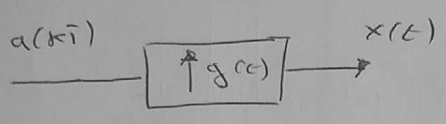
$$X(f) = 3T \text{sinc}\left(\frac{3T k}{5T}\right) = 3T \text{sinc}\left(\frac{3}{5} k\right)$$

Trasformata di Fourier di:

$$x(t) = \sum_{k=-\infty}^{+\infty} p^k \text{rect}\left(\frac{2(t-kT)}{T}\right) \quad |p| < 1$$

$x(t)$ può essere interpretato come l'uscita di un filtro interpolatore $Z(T) \rightarrow \mathbb{R}$

come segnale di ingresso abbiamo $a(kT) = \frac{p^k}{T} 1_0(kT)$



dove $g(t) = \text{rect}\left(\frac{2t}{T}\right) \rightarrow \text{per } k=0$

$$X(f) = A(f) G(f)$$

$$A(f) = T \sum_{k=-\infty}^{+\infty} a(kT) e^{-j2\pi f kT} = T \sum_{k=0}^{+\infty} \frac{p^k}{T} e^{-j2\pi f kT} = \frac{1}{1 - p e^{-j2\pi f T}}$$

$$G(f) = \frac{T}{2} \text{sinc}\left(\frac{T}{2} f\right)$$

$$X(f) = \frac{1}{1 - p e^{-j2\pi f T}} \frac{T}{2} \text{sinc}\left(\frac{T}{2} f\right)$$

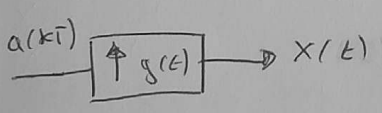
Trasformata di Fourier di:

$$x(t) = \sum_{k=-\infty}^{+\infty} 0,1^{|k|} \text{tri}\left(\frac{4(t-kT)}{T}\right)$$

$x(t)$ può essere interpretato come l'uscita di un filtro interpolatore $Z(T) \rightarrow \mathbb{R}$

come segnale di ingresso abbiamo $a(kT) = 0,1^{|k|}$

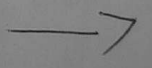
$$x(t) = T \sum_{k=-\infty}^{+\infty} 0,1^{|k|} \cdot \frac{1}{T} \text{tri}\left(\frac{4(t-kT)}{T}\right)$$



dove $g(t) = \frac{1}{T} \text{tri}\left(\frac{4t}{T}\right) \rightarrow \text{per } k=0$

$$A(f) = T \sum_{k=-\infty}^{+\infty} 0,1^{|k|} e^{-j2\pi f kT} = T \left(\frac{1}{1 - 0,1 e^{-j2\pi f T}} + \frac{1}{1 - 0,1 e^{j2\pi f T}} - 1 \right)$$

$$G(f) = \frac{1}{4} \cdot \frac{1}{T} \text{sinc}^2\left(\frac{T}{4} f\right) = \frac{1}{4} \text{sinc}^2\left(\frac{T}{4} f\right)$$



$$X(f) = A(f) G(f) = \frac{T}{4} \left(\frac{1}{1 - 0,1 e^{-j2\pi f T}} + \frac{1}{1 - 0,1 e^{+j2\pi f T}} - 1 \right) \text{sinc}^2 \left(\frac{T}{4} f \right)$$

Calcolo della trasformata di Fourier di:

$$x(kT) = 0,1^{|k|} + 3 \cdot 0,1^{|k-3|}$$

$$x_1(kT) = 0,1^{|k|}$$

$$X(f) = X_1(f) + 3 X_1(f) e^{-j2\pi 3f}$$

$$X_1(f) = T \sum_{k=-\infty}^{+\infty} 0,1^{|k|} e^{-j2\pi f k T} = T \sum_{k=0}^{+\infty} 0,1^k e^{-j2\pi f k T} + T \sum_{k=0}^{+\infty} 0,1^k e^{+j2\pi f k T} - T =$$

$$= T \left(\frac{1}{1 - 0,1 e^{-j2\pi f T}} + \frac{1}{1 - 0,1 e^{+j2\pi f T}} - 1 \right)$$

$$X(f) = X_1(f) + 3 X_1(f) e^{-j2\pi 3f}$$

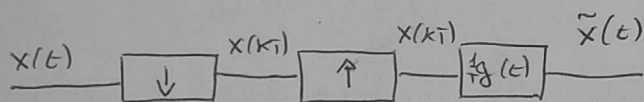
Calcolo della trasformata di Fourier di:

$$\tilde{x}(t) = \sum_{k=-\infty}^{+\infty} x(kT) \text{rect} \left(\frac{t - kT}{T} \right) \quad t \in \mathbb{R}$$

$$x(t) = V_0 \text{sinc}^2 \left(\frac{t}{T_1} \right)$$

$$T_c = \frac{2}{T_1}$$

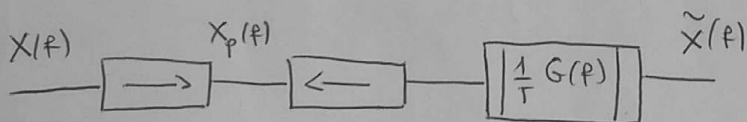
$$T = \frac{1}{T_c} = T_p$$



$$g(t) = \text{rect} \left(\frac{t}{T} \right)$$

$x(t)$ viene campionato e quindi interpolato per poi essere filtrato da $\frac{1}{T} g(t)$

Passando in frequenza abbiamo



$$X(f) = \mathcal{F} \{ x(t) \} = T_1 V_0 \text{tri} \left(f T_1 \right)$$

$$G(f) = \mathcal{F} \{ g(t) \} = T \text{sinc} \left(f T \right)$$

$$X_p(f) = \sum_{k=-\infty}^{+\infty} X \left(f - \frac{k}{T_p} \right) = \sum_{k=-\infty}^{+\infty} T_1 V_0 \text{tri} \left[\left(f - \frac{k}{T_p} \right) T_1 \right]$$

L'effetto della deperiodizzazione non cambia il segnale che di fatto rimane lo stesso

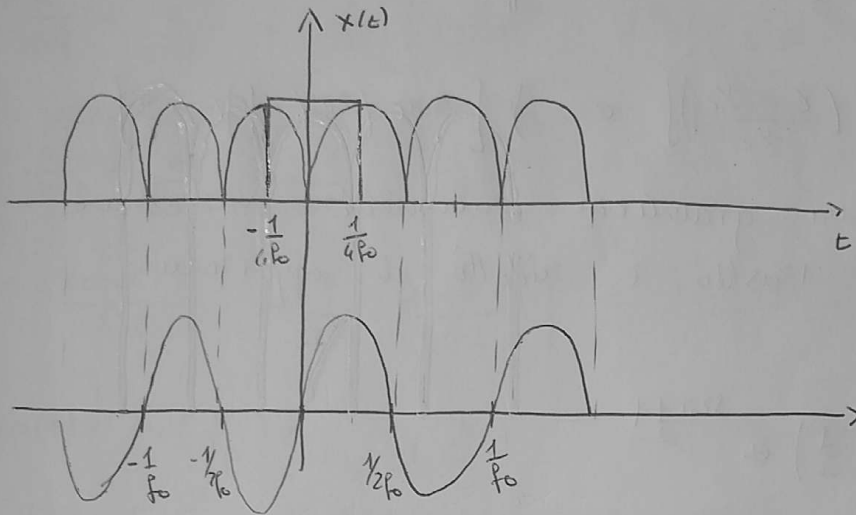


$$\tilde{X}(f) = x_p(f) \cdot \frac{1}{T} G(f)$$

$$= \text{sinc}(fT) \left[\sum_{k=-\infty}^{+\infty} T_1 V_0 \text{tri} \left(\left(f - \frac{k}{T_p} \right) T_1 \right) \right]$$

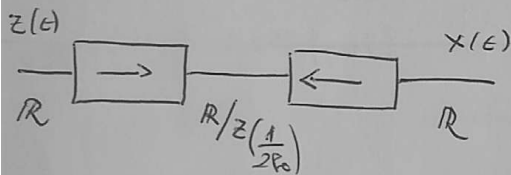
Calcolo della trasformata di Fourier di:

$$x(t) = |\sin 2\pi f_0 t|$$

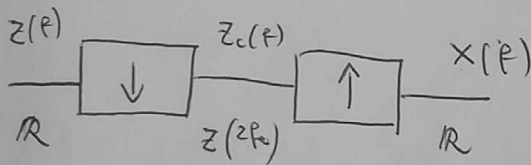


Per costruire $|\sin 2\pi f_0 t|$ prendo un rect di ampiezza $\frac{1}{2f_0}$ e periodizzo tale segnale:

$$z(t) = \sin(2\pi f_0 t) \text{rect} \left(\frac{t}{\frac{1}{2f_0}} \right)$$



↓ Passando in frequenza



$$x(f) = \sum_{k=-\infty}^{+\infty} x\left(\frac{k}{T_p}\right) \delta\left(f - \frac{k}{T_p}\right)$$

$$Z(f) = \left\{ \frac{1}{2f_0} \text{sinc} \left(\frac{f}{2f_0} \right) \right\} * \left\{ \frac{j}{2} \delta(f+f_0) - \frac{j}{2} \delta(f-f_0) \right\}$$

$$= \frac{1}{2f_0} \cdot \frac{j}{2} \left[\text{sinc} \left(\frac{f+f_0}{2f_0} \right) - \text{sinc} \left(\frac{f-f_0}{2f_0} \right) \right]$$

$$Z_c(f) = X(f) = \sum_{k=-\infty}^{+\infty} \frac{j}{4f_0} \left[\text{sinc} \left(\frac{f+f_0 - \frac{k}{T_p}}{2f_0} \right) - \text{sinc} \left(\frac{f-f_0 - \frac{k}{T_p}}{2f_0} \right) \right]$$

$$T_p = \frac{1}{2f_0}$$

Trasformata di Fourier di:

$$x(t) = \text{tri}\left(-2t + 1\right) \sin\left(-2t + \frac{\pi}{6}\right)$$

$$x(t) = \text{tri}\left(\frac{t - \frac{1}{2}}{\frac{1}{2}}\right) \sin\left(-2t + \frac{\pi}{6}\right)$$

$$\sin\left(-2t + \frac{\pi}{6}\right) = \sin\left(2\pi\left(-\frac{1}{\pi}\right)t + \frac{\pi}{6}\right) \quad f_0 = -\frac{1}{\pi}$$

$$X(f) \Rightarrow \text{condizione per } \mathcal{F}\left[\text{tri}\left(\frac{t - \frac{1}{2}}{\frac{1}{2}}\right)\right] \text{ e } \mathcal{F}\left[\sin\left(2\pi\left(-\frac{1}{\pi}\right)t + \frac{\pi}{6}\right)\right]$$

combinando in seguito per una grandezza sinusoidale si ottiene lo stesso segnale sdoppiato, traslato e ridotto di opportuni coefficienti

$$\mathcal{F}\left\{\text{tri}\left(\frac{t - \frac{1}{2}}{\frac{1}{2}}\right)\right\} = \frac{1}{2} \text{sinc}^2\left(\frac{f}{2}\right) e^{-j2\pi \frac{1}{2}f}$$

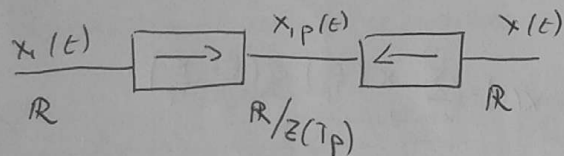
$$\mathcal{F}\left\{\sin\left(2\pi\left(-\frac{1}{\pi}\right)t + \frac{\pi}{6}\right)\right\} = \frac{j}{2} \delta\left(f - \frac{1}{\pi}\right) e^{-\frac{j\pi}{6}} + \frac{j}{2} \delta\left(f + \frac{1}{\pi}\right) e^{\frac{j\pi}{6}}$$

$$X(f) = \frac{j}{4} \text{sinc}^2\left(\frac{1}{2}\left(f - \frac{1}{\pi}\right)\right) e^{-j2\pi \frac{1}{2}\left(f - \frac{1}{\pi}\right) - \frac{j\pi}{6}} + \frac{j}{4} \text{sinc}^2\left(\frac{1}{2}\left(f + \frac{1}{\pi}\right)\right) e^{-j2\pi \frac{1}{2}\left(f + \frac{1}{\pi}\right) + \frac{j\pi}{6}}$$

Trasformata di Fourier di:

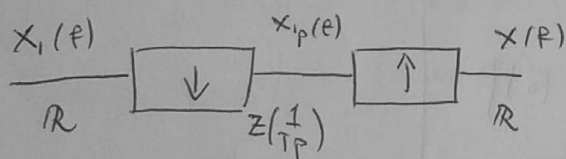
$$x(t) = \sum_{k=-\infty}^{+\infty} (t - 2k) \text{rect}\left(\frac{t - 2k}{2}\right)$$

Questo segnale si tratta di una periodizzazione di $t \text{rect}\left(\frac{t}{2}\right) = x_1(t)$



$$T_p = 2$$

↓ In Frequenza



$$X_1(f) = \int_{-\infty}^{+\infty} t \text{rect}\left(\frac{t}{2}\right) e^{-j2\pi f t} dt = \int_{-1}^1 t e^{-j2\pi f t} dt \rightarrow \text{integro per parti} \rightarrow$$

$$\begin{aligned}
 f &= t & f' &= 1 \\
 g &= -\frac{1}{520f} e^{-520ft} & g' &= e^{-520ft} \Rightarrow \int_{-1}^1 \frac{-t}{520f} e^{-520ft} dt + \int_{-1}^1 \frac{1}{520f} e^{-520ft} dt = \\
 &= -\frac{e^{-520f}}{520f} - \left(\frac{e^{520f}}{520f} \right) + \frac{1}{520f} \left(-\frac{1}{520f} e^{-520f} \right) - \left(-\frac{1}{520f} e^{-520f} \right) = \\
 &= -\frac{e^{-520f}}{520f} - \frac{e^{520f}}{520f} + \frac{1}{520f} \left[\left(-\frac{1}{520f} e^{520f} \right) - \left(-\frac{1}{520f} e^{-520f} \right) \right] = \\
 &= -\frac{e^{-520f}}{520f} - \frac{e^{520f}}{520f} + \frac{1}{520f} \left(-\frac{1}{520f} e^{520f} + \frac{e^{-520f}}{520f} \right) = \\
 &= -\frac{\cos 2\pi f}{520f} + \frac{1}{520f} \left(\frac{\sin 2\pi f}{\pi f} \right) = j \frac{\cos 2\pi f}{\pi f} - \frac{j}{2\pi^2 f^2} \sin(2\pi f)
 \end{aligned}$$

$$x_p(f) = X_1\left(\frac{k}{T_p}\right) = X_1\left(\frac{k}{2}\right) \rightarrow \text{campioneamento}$$

$$\begin{aligned}
 X(f) &= \frac{1}{T_p} \sum_{k=-\infty}^{+\infty} x_p\left(\frac{k}{2}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{T_p}\right) \\
 &= \frac{1}{2} \sum_{k=-\infty}^{+\infty} x_p\left(\frac{k}{2}\right) \delta_{\mathbb{R}}\left(f - \frac{k}{2}\right)
 \end{aligned}$$

Trasformata di Fourier di:

$$x(nT) = \begin{cases} \cos(2\pi f_0 nT) & n = 0, 1, \dots, N-1 \\ 0 & \text{altrove} \end{cases}$$

$$\begin{aligned}
 X(nT) &= S(nT) P(nT) & \text{con } S(nT) &= \cos(2\pi f_0 nT) \\
 X(f) &= S(f) * P(f) & p(nT) &= \begin{cases} 1 & n = 0, \dots, N-1 \\ 0 & \text{altrove} \end{cases}
 \end{aligned}$$

$$S(f) = \frac{1}{2} \delta_{\frac{2(f-f_0)}{T}} + \frac{1}{2} \delta_{\frac{2(f+f_0)}{T}} \rightarrow \text{in frequenza periodizzazio}$$

$$P(f) = NT \operatorname{sinc}_N\left(\frac{N-1}{2}\right) e^{-j2\pi\left(\frac{N-1}{2}\right) fT}$$

$$X(f) = \frac{NT}{2} \left[\operatorname{sinc}_N\left((f-f_0)NT\right) e^{-j2\pi\left(\frac{N-1}{2}\right)(f-f_0)T} + \operatorname{sinc}_N\left((f+f_0)NT\right) e^{-j2\pi\left(\frac{N-1}{2}\right)(f+f_0)T} \right]$$

Calcolo trasformata di Fourier di:

$$x(t) = \text{tri}(-t+1) \cos(-2t + \frac{\pi}{3})$$

$$= \text{tri}(t-1) \cos(2t - \frac{\pi}{3})$$

$$= \underbrace{\text{tri}(t-1)}_{x_1(t)} \underbrace{\cos(2\pi \frac{1}{11} t - \frac{\pi}{3})}_{x_2(t)} \rightarrow f_0 = \frac{1}{11}$$

$$x_1(f) = \mathcal{F}[\text{tri}(t-1)] = \text{sinc}^2(f) e^{-j2\pi f}$$

$$x_2(f) = \mathcal{F}[\cos(2\pi \frac{1}{11} t - \frac{\pi}{3})] = \frac{1}{2} \delta(f - \frac{1}{11}) e^{-j\frac{\pi}{3}} + \frac{1}{2} \delta(f + \frac{1}{11}) e^{j\frac{\pi}{3}}$$

$$X(f) = x_1(f) * x_2(f) = \frac{1}{2} e^{-j\frac{\pi}{3}} \text{sinc}^2(f - \frac{1}{11}) e^{-j2\pi(f - \frac{1}{11})} + \frac{1}{2} e^{j\frac{\pi}{3}} \text{sinc}^2(f + \frac{1}{11}) e^{-j2\pi(f + \frac{1}{11})}$$

Calcolo trasformata di Fourier di:

$$x(t) = \sum_{k=0}^{+\infty} 0.5^k \text{sinc}^2\left(\frac{t-kT}{T}\right)$$

$x(t)$ può essere considerato come l'uscita di un filtro interpolatore:

$$\frac{a(kT)}{z(T)} \rightarrow \boxed{\uparrow g(t)} \rightarrow \frac{x(t)}{R}$$

$$a(kT) = 0.5^k 1_0(kT)$$

$$g(t) = \frac{1}{T} \text{sinc}^2\left(\frac{t}{T}\right)$$

$$X(f) = A(f) G(f)$$

$$A(f) = T \sum_{k=0}^{+\infty} 0.5^k e^{-j2\pi f k T} = \frac{T}{1 - 0.5 e^{-j2\pi f T}}$$

$$G(f) = \frac{1}{T} \text{tri}(fT) = \text{tri}(fT)$$

$$X(f) = \frac{T}{1 - 0.5 e^{-j2\pi f T}} \text{tri}(fT)$$

Calcolo trasformata di Fourier di:

$$x(kT) = -k 0,5^{-k} 1_0(-kT)$$

$$X(f) = T \sum_{k=-\infty}^{+\infty} x(kT) e^{-j2\pi f kT}$$

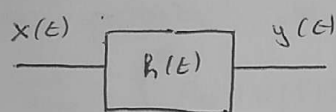
$$\sum_{n=0}^{+\infty} n p^{n-1} = \frac{1}{(1-p)^2}$$

$$X(f) = T \sum_{k=-\infty}^{+\infty} -k 0,5^k 1_0(-kT) e^{-j2\pi f kT} = \sum_{k=-\infty}^0 -k 0,5^k e^{-j2\pi f kT} =$$

$$= \sum_{k=0}^{+\infty} k 0,5^k e^{-j2\pi f kT} = \sum_{k=0}^{+\infty} k 0,5^{(k-1)} \cdot 0,5 \cdot e^{-j2\pi f (k-1)T} e^{-j2\pi f T} =$$

$$= T 0,5 e^{j2\pi f T} \cdot \sum_{k=0}^{+\infty} k (0,5 e^{-j2\pi f T})^{(k-1)} = \frac{T \cdot 0,5 e^{j2\pi f T}}{(1 - 0,5 e^{-j2\pi f T})^2}$$

Calcolo uscita filtro



$$x(t) = v_0 + v_1 \sin\left(\frac{3\pi t}{T}\right)$$

$$h(t) = h_0 \operatorname{sinc}^3\left(\frac{t}{T}\right)$$

$$y(t) = ?$$

calcolo della potenza di $y(t) = ?$

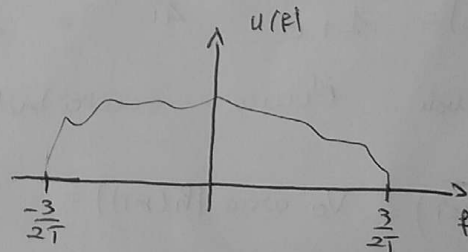
$$y(t) = x * h(t)$$

$U(f)$ è la convoluzione di 3 rect con banda pari a:

$$\operatorname{sinc}\left(\frac{t}{T}\right) \leftrightarrow T \operatorname{rect}(fT) \Rightarrow B = \frac{1}{2T}$$

La banda totale di $U(f)$ sarà quindi

$$\frac{3}{2T}$$



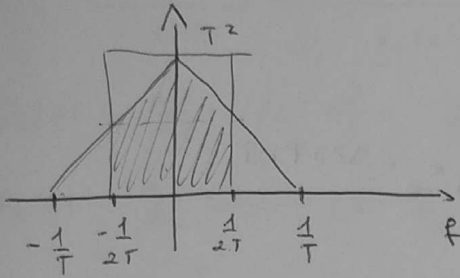
$$v_1 \sin\left(\frac{3\pi t}{T}\right) \Rightarrow f_0 = \frac{3}{2T}$$

Quindi l'uscita della componente $\sin\left(\frac{3\pi t}{T}\right)$ è nulla, l'unica componente che sopravvive è la costante $v_0 \Rightarrow x(t) = v_0$

$$y(t) = V_0 \text{ area}(R(t)) = V_0 u(0)$$

area $R(t)$:

$$h_0 \int_{-\infty}^{+\infty} \text{sinc}^3\left(\frac{t}{T}\right) dt = h_0 \int_{-\infty}^{+\infty} \text{sinc}^2\left(\frac{t}{T}\right) \text{sinc}\left(\frac{t}{T}\right) dt \stackrel{\text{Parseval}}{=} h_0 \int_{-\infty}^{+\infty} T \text{Tri}(fT) \cdot T \text{rect}(fT) df$$



$$\text{Area} = h_0 \left(\frac{1}{T} \cdot \frac{T^2}{2} + \frac{1}{T} \cdot \frac{T^2}{2} \cdot \frac{1}{2} \right) = h_0 \left(\frac{T}{2} + \frac{T}{4} \right) = h_0 \left(\frac{2T+1T}{4} \right) = \frac{h_0 3T}{4}$$

$$P_y = \lim_{T \rightarrow +\infty} \int_{-T}^T |y(t)|^2 dt = \frac{h_0^2 9T^2}{4^2}$$

Filtro a tempo discreto $u_z(z) = 1 + z^{-2}$

$$\text{Ingresso } x(kT) = V_0 + \sin\left(k\frac{\pi}{2} + \frac{\pi}{10}\right)$$

$$x_0(kT) = x(kT) \delta_0(kT)$$

$$H(f) = 1 + e^{-j2\pi 2fT}$$

Analizziamo il segnale $x(kT)$

$V_0 \rightarrow$ costante

$$\sin\left(k\frac{\pi}{2} + \frac{\pi}{10}\right) = \sin\left(2\pi f_0 kT + \frac{\pi}{10}\right) \Rightarrow f_0 = \frac{1}{4T}$$

$$u(f_0) = 1 + e^{-j4\pi \frac{1}{4T} T} = 1 + e^{-j\pi} = 1 - 1 = 0$$

Quindi l'unico contributo all'uscita del filtro è V_0

$$y(kT) = V_0 \text{ area}(R(kT)) = V_0 u(0) = 2V_0$$

③ Sistema di trasmissione PAM simboli equiprobabili $a_k = \pm 2$
 periodo di simbolo T .

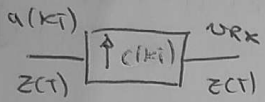
$$g(t) = V_0 \operatorname{sinc}\left(\frac{t}{T}\right) \operatorname{sinc}\left(\frac{t}{2T}\right) \quad \text{impulso in trasmissione}$$

$$L(f) = A_m \quad \text{sulla banda di } g(t)$$

$$H(f) = A_R \operatorname{rect}\left(\frac{2Tf}{3}\right)$$

$$AWGN \rightarrow R_0$$

Probabilità di errore = ?



La banda di $G(f)$ è data dalla convoluzione di $T \operatorname{rect} fT$ e $2T \operatorname{rect} 2fT$

$$B = \frac{1}{2T} + \frac{1}{4T} = \frac{2+1}{4} T = \frac{3}{4} T$$

Tale banda è la stessa per $L(f)$ ed $u(f)$

$$\text{Quindi } c(t) = A_m A_R g(t) \quad c(kT) = V_0 A_m A_R T \delta_{2T}(kT)$$

La potenza del rumore è data da:

$$\sigma_w^2 = \int_{-\infty}^{+\infty} R_0 |u(f)|^2 df = 2 \int_0^{\frac{3}{4}T} A_R^2 R_0 df = 2 \cdot \frac{3}{4} T A_R^2 R_0 = \frac{A_R^2 R_0 T}{2}$$

$$P_e = Q\left(\frac{2V_0'}{\sigma_w}\right) \quad V_0' = c(0) = V_0 A_m A_R$$

È il fattore 2 al numeratore dovuto al fatto che $a_k = \pm 2$

Esame 23/06/2004

①

$a(kT) \rightarrow$ processo aleatorio gaussiano a tempo discreto

$\mu_a = 0 \quad \sigma_a^2 = \sigma_a^2$

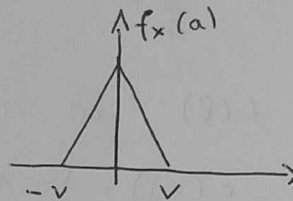
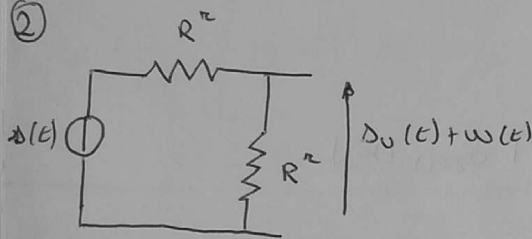
Calcolare $P[a(kT) + a(kT-T) > 1] = P[y > 1] = Q\left(\frac{1 - \mu_y}{\sigma_y}\right)$

$\mu_y = E[y] = E[a(kT) + a(kT-T)] = E[a(kT)] + E[a(kT-T)] = \mu_a + \mu_a = 0$

$\sigma_y^2 = E[(y - \mu_y)^2] = E[y^2] = E[(a(kT) + a(kT-T))^2] = E[a^2(kT) + a^2(kT-T) + 2a(kT)a(kT-T)]$
 $= E[a^2(kT)] + E[a^2(kT-T)] = 2\sigma_a^2$

$P[y > 1] = Q\left(\frac{1}{\sqrt{2}\sigma_a}\right)$

②



$\rightarrow$ Densità di probabilità di $s(t)$

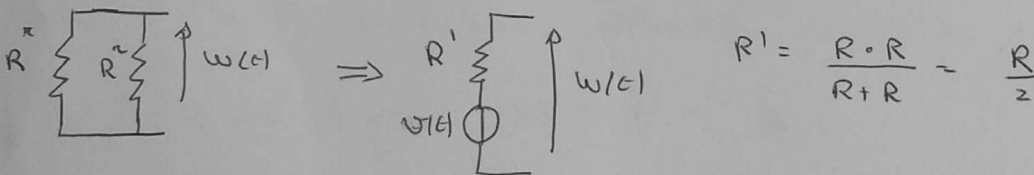
$w \rightarrow$ è dato dal rumore termico delle resistenze poste entrambe a temperatura T_0

calcolare $SNR = \frac{M_{s_u}}{M_w}$

$s_u(t) = s(t) \cdot \frac{R}{R+R} = \frac{s(t)}{2}$

$M_{s_u} = E[s_u^2] = \frac{1}{4} M_s = \frac{1}{4} \left[2 \int_0^V a^2 f_x(a) da \right] = \frac{1}{2} \int_0^V a^2 \left(\frac{1}{V} - \frac{1}{V^2} a \right) da = \frac{1}{2} \left[\frac{a^3}{3V} - \frac{a^4}{4V^2} \right]_0^V = \frac{1}{2} \left(\frac{V^3}{3} - \frac{V^4}{4V} \right) = \frac{1}{2} \left(\frac{V^3}{3} - \frac{V^3}{4} \right) = \frac{1}{2} \cdot \frac{V^3}{12} = \frac{V^3}{24}$

Per quanto riguarda il calcolo di $M_w = E[w^2]$



Utilizzando il teo di Nyquist: $R_w(f) = 2kT_0 \frac{R}{2} |f| = kT_0 R$

$M_w = \int_{-B}^B kT_0 R |f| df = 2kT_0 R B$

$SNR = \frac{E[s_u(t)^2]}{E[w(t)^2]} = \frac{V^2}{48 kT_0 R B}$

$$h(t) = h_0 \operatorname{sinc}^2\left(\frac{4t}{T}\right)$$

$$x(t) = \cos\left(2\pi \frac{t}{T} + \frac{\pi}{4}\right) \quad t \in \mathbb{R}$$

calcolare l'uscita del filtro:

$$y(t) = A_0 |H(f_0)| \cos\left(2\pi f_0 t + \varphi + \angle H(f_0)\right)$$

$$\cos\left(2\pi f_0 t + \varphi\right) \Leftrightarrow \cos\left(2\pi \frac{t}{T} + \frac{\pi}{4}\right) \Rightarrow f_0 = \frac{1}{T}$$

$$H(f) = h_0 \frac{T}{4} + j \left(\frac{T}{4} f \right)$$

$$H(f_0) = h_0 \frac{T}{4} + j \left(\frac{T}{4} \frac{1}{T} \right) = h_0 \frac{T}{4} + j \left(\frac{1}{4} \right)$$

$$y(t) = h_0 \frac{T}{4} + j \left(\frac{1}{4} \right) \cos\left(2\pi \frac{t}{T} + \frac{\pi}{4}\right)$$

$$x(kT) = (-1)^k = e^{jk\pi} = \cos(k\pi) \quad \text{con } kT \in \mathbb{Z}(T)$$

$$h(kT) = (-1)^k \operatorname{rect} \frac{kT}{5T}$$

$$x(kT) = \cos(k\pi) \rightarrow \cos\left(2\pi f_0 kT\right) = \cos\left(2\pi \frac{1}{2T} kT\right) \quad f_0 = \frac{1}{2T}$$

Trovare per calcolo dell'uscita al ingresso discretizzato:

$$y(kT) = A_0 |H(f_0)| \cos\left(2\pi f_0 kT + \varphi + \angle H(f_0)\right)$$

$$u(f) = \mathcal{F}\left[(-1)^k\right] * \mathcal{F}\left[\operatorname{rect} \frac{kT}{5T}\right] = \mathcal{F}\left[e^{jk\pi}\right] * \mathcal{F}\left[\operatorname{rect} \frac{kT}{5T}\right] =$$

$$= \mathcal{F}\left[e^{j2\pi \frac{1}{2T} kT}\right] * (5T \operatorname{sinc}_s(5fT)) = \delta\left(f - \frac{1}{2T}\right) * (5T \operatorname{sinc}_s(5fT)) =$$

$$= 5T \operatorname{sinc}_s\left(5T\left(f - \frac{1}{2T}\right)\right)$$

$$x(t) = (1 + \sin^2(t)) \cos(2\pi f_0 t) \quad f_0 > 2$$

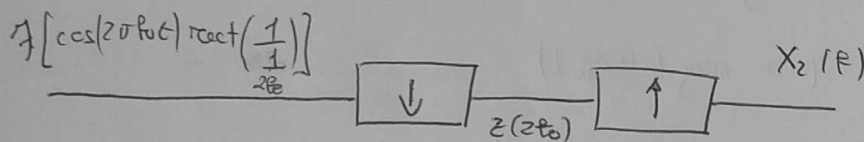
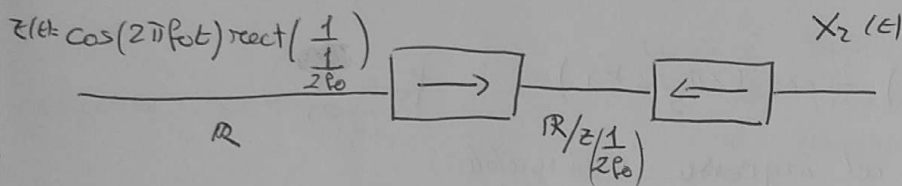
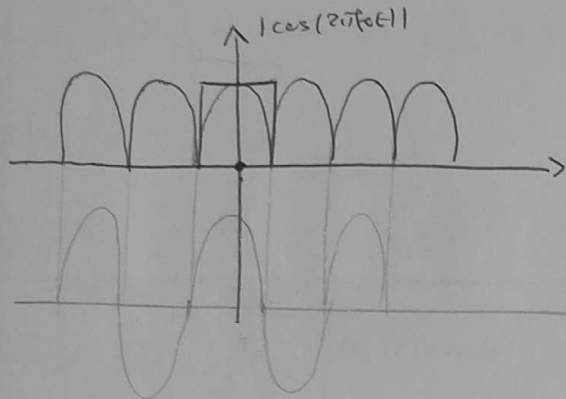
$$h(t) = 2 \operatorname{sinc}(2t) \quad y(t) = ?$$

$$x(t) = (1 + \sin^2(t)) \cdot |\cos(2\pi f_0 t)| = x_1(t) \cdot x_2(t)$$

$$X(f) = X_1(f) * X_2(f)$$

$$X_1(f) = \mathcal{F}[1 + \sin^2(t)] = \delta_R + \tau_1(f)$$

$$X_2(f) = \mathcal{F}[|\cos(2\pi f_0 t)|]$$



$$z(f) = \left[\frac{1}{2f_0} \operatorname{sinc}\left(\frac{f-f_0}{2f_0}\right) + \frac{1}{2f_0} \operatorname{sinc}\left(\frac{f+f_0}{2f_0}\right) \right] \cdot \frac{1}{2}$$

$$= \frac{1}{4f_0} \operatorname{sinc}\left(\frac{f-f_0}{2f_0}\right) + \frac{1}{4f_0} \operatorname{sinc}\left(\frac{f+f_0}{2f_0}\right)$$

$$X_2(f) = 2f_0 \sum_{k=-\infty}^{+\infty} z(2kf_0) \delta_R(f - 2kf_0)$$

$$X(f) = 2f_0 \sum_{k=-\infty}^{+\infty} z(2kf_0) X_1(f - 2kf_0)$$

Dopo il filtraggio con $H(f) = \operatorname{rect}\left(\frac{f}{2}\right)$ resta $Y(f) = 2f_0 z(0) X_1(f)$

$$y(t) = \frac{2}{\pi} (1 + \sin^2(t))$$

Filtro a tempo discreto:

Calcolare la risposta impulsiva del filtro:

$$H(f) = \frac{T}{1 - 0,4e^{-j2\pi fT} + 0,03e^{-j4\pi fT}} \quad f \in \mathbb{R}/\mathbb{Z}\left(\frac{1}{T}\right)$$

$$H(z) = \frac{T}{1 - 0,4z^{-1} + 0,03z^{-2}}$$

calcoliamo i poli $z^2 - 0,4z + 0,03 = 0$ $z_{1,2} = \frac{0,4 \pm \sqrt{(0,4)^2 - 4 \cdot 0,03}}{2}$ $\begin{matrix} 0,1 \\ 0,3 \end{matrix}$

$$H(z) = \frac{T}{(1 - 0,1z^{-1})(1 - 0,3z^{-1})}$$
$$= \frac{A}{1 - 0,1z^{-1}} + \frac{B}{1 - 0,3z^{-1}}$$

$$A = \left(\cancel{1 - 0,1z^{-1}} \right) \frac{T}{(1 - \cancel{0,1z^{-1}})(1 - 0,3z^{-1})} \Big|_{z=0,1} = -\frac{1}{2}T$$

$$B = \left(1 - \cancel{0,3z^{-1}} \right) \frac{T}{(1 - 0,1z^{-1})(1 - \cancel{0,3z^{-1}})} \Big|_{z=0,3} = \frac{3}{2}T$$

$$H(z) = \frac{-\frac{1}{2}T}{1 - 0,1z^{-1}} + \frac{\frac{3}{2}T}{1 - 0,3z^{-1}}$$

$$h(kT) = -\frac{1}{2}T(0,1)^k 1_0(kT) + \frac{3}{2}T(0,3)^k 1_0(kT)$$

Filtro a tempo discreto

$$a(kT) \xleftrightarrow{f} A(f) \quad f \in \mathbb{R}/\mathbb{Z}(\frac{1}{T})$$

$$b(kT) = 0,5 a(kT) + 0,5 a(kT-T)$$

$$B(f) = ?$$

$$B(f) = 0,5 A(f) + 0,5 A(f) e^{-j2\pi f T}$$

$$B(f) = 0,5 A(f) [1 + e^{-j2\pi f T}]$$

$$a(kT) \xleftrightarrow{f} A(f) \quad f \in \mathbb{R}/\mathbb{Z}(\frac{1}{T})$$

$$b(kT) = a(kT) + 0,9 b(kT-T)$$

$$B(f) = ?$$

$$B(f) = A(f) + 0,9 B(f) e^{-j2\pi f T}$$

$$B(f) = \frac{A(f)}{1 - 0,9 e^{-j2\pi f T}}$$

$$a(kT) = \cos(2\pi f_0 kT) \quad f_0 = \frac{1}{8T}$$

$$b(kT) = a(kT) + 0,5 b(kT-T)$$

$$b(kT) = ?$$

$$B(f) = \frac{A(f)}{1 - 0,5 e^{-j2\pi f T}}$$

essendo $a(kT)$ un ingresso di tipo sinusoidale, la risposta sarà:

$$b(kT) = |H(f_0)| \cos(2\pi f_0 kT + \arg(H(f_0)))$$

$$\text{con } H(f_0) = \frac{1}{1 - 0,5 e^{-j2\pi \frac{1}{8T} T}}$$

$$H(z) = \frac{1}{1 - 0,4 z^{-1} + 0,03 z^{-2}} \quad f \in \mathbb{R}/\mathbb{Z}(\frac{1}{T})$$

$$P(z) \rightarrow z^2 - 0,4z + 0,03 \Rightarrow z_{1,2} = \frac{0,4 \pm \sqrt{0,16 - 0,12}}{2} = \frac{0,4 \pm 0,2}{2} \begin{matrix} / 0,1 \\ \backslash 0,3 \end{matrix}$$

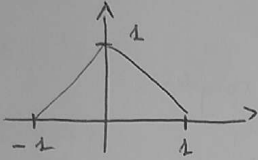
Esercizio 2

Energia del segnale a tempo continuo e sua trasformata

$$x(t) = \text{sinc}(t+1) + \text{sinc}(t-1)$$

$$E_x = \int_{-\infty}^{+\infty} |x(t)|^2 dt = \int_{-\infty}^{+\infty} |2\text{sinc}(t)|^2 dt = \int_{-\infty}^{+\infty} 4 \text{tri}(f) df = 4$$

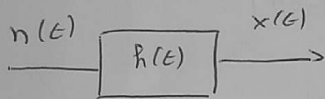
↑
Parseval



$$A = \frac{2 \cdot 1}{2} = 1$$

Esercizio 3

Processo aleatorio stocastico $x(t)$ $t \in \mathbb{R}$



$$R_n(f) = R_0$$

$$h(t) = g * g_-(t) = R_0 \text{sinc}^2\left(\frac{t}{T}\right)$$

$$P\left[x(t) < x\left(t + \frac{T}{2}\right)\right]$$

$$P\left[x(t) - x\left(t + \frac{T}{2}\right) < 0\right] = P[Y < 0] = \Phi\left(\frac{0 - m_y}{\sigma_y}\right)$$

$$m_y = E[Y] = E\left[x(t) - x\left(t + \frac{T}{2}\right)\right]$$

$$m_x = m_n \cdot \text{area } n(t) = m_n \cdot H(0)$$

essendo $n(t)$ univoco bianco gaussiano $m_n = 0 \Rightarrow m_x = 0$

$$\text{da cui } m_y = 0 \quad P[Y < 0] = \Phi(0) = \frac{1}{2}$$

Esercizio 1

$$U_z(z) = \frac{T}{1 + z^{-1} + 0.5z^{-2}}$$

$$h(kT) = ?$$

$$h(kT) = \sum_{n=1}^q A_n P_n^k (10(kT)) \rightarrow \text{se i poli sono reali}$$

$$h(kT) = 2 \operatorname{Re} (A_{n0} |P_{n0}|^k e^{j k \arg(P_{n0})} 10(kT)) \rightarrow \text{se i poli sono complessi coniugati}$$

Determiniamo i poli:

$$z^2 + z + 0.5 = 0 \quad z_{1,2} = \frac{-1 \pm \sqrt{1-2}}{2} = \frac{-1 \pm i}{2} \quad \begin{matrix} \nearrow -\frac{1+i}{2} \\ \searrow -\frac{1-i}{2} \end{matrix}$$

Determiniamo i residui (saranno anch'essi complessi coniugati):

$$U_z(z) = \frac{A}{1 - p_1 z^{-1}} + \frac{B}{1 - p_2 z^{-1}}$$

$$A = \left(1 - \left(-\frac{1-i}{2} \right) z \right) \frac{1}{\left(1 - \left(-\frac{1+i}{2} \right) z \right) \left(1 - \left(-\frac{1-i}{2} \right) z \right)} \bigg|_{z = -\frac{1+i}{2}} = \frac{1}{1 - \left(-\frac{1-i}{2} \right) \left(-\frac{1+i}{2} \right)} =$$

$$= \frac{1}{1 - \left(-\frac{1-i}{2} \right) \left(\frac{2}{-1+i} \right)} = \frac{1}{1 - \left(\frac{-1-i}{-1+i} \right)} = \frac{1}{1 - \left(\frac{-1-i}{-1+i} \right) \left(\frac{-1-i}{-1-i} \right)} = \frac{1}{1 - \left(\frac{1+i+i-1}{1+1} \right)} =$$

$$= \frac{1}{1 - \frac{2i}{2}} = \frac{1}{1-i} \frac{(1+i)}{(1+i)} = \frac{1+i}{1+1} = \frac{1+i}{2}$$

$$B = A^* = \frac{1-i}{2}$$

Prendendo i residui complessi coniugati:

$$h(kT) = 2 \operatorname{Re} \left(\frac{1+i}{2} \left| -\frac{1+i}{2} \right|^k e^{j k \arg \left(-\frac{1+i}{2} \right)} 10(kT) \right)$$

$$= 2 \left(\frac{1}{2} \left(\frac{1}{2} \right)^k e^{-j \frac{\pi}{4} k} 10(kT) \right) = \left(\frac{1}{2} \right)^k e^{-j \frac{\pi}{4} k} 10(kT)$$

$$\arg \left(-\frac{1+i}{2} \right) = \arctan \left(\frac{\frac{1}{2}}{-\frac{1}{2}} \right) = \frac{\pi}{4}$$

PAM

$$a(kT) = \{0, \pm 1\} \quad M=3 \quad P_e = \frac{2M-2}{M} Q\left(\frac{V_0'}{2\sigma_n}\right)$$

$$V_0' = c(0) \quad \sigma_n^2 = R_0 \int_{-\infty}^{\infty} |h(t)|^2 dt = R_0 \int_{-\infty}^{\infty} |u(f)|^2 df$$

$$C(f) = G(f) U(f) L(f)$$

$$g(t) = V_0 \operatorname{tri}\left(\frac{2t}{T}\right) \longleftrightarrow G(f) = V_0 \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2} f\right)$$

$$L(f) = A_M$$

progetto con matched filter:

$$h(t) = \kappa \tilde{g}(-t)$$

$$\tilde{g} = g * l(t)$$

$$\tilde{G}(f) = G(f) L(f) = A_M V_0 \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2} f\right)$$

$$\tilde{g}(t) = A_M V_0 \frac{T}{2} \operatorname{tri}\left(\frac{2t}{T}\right) = A_M V_0 \operatorname{tri}\left(\frac{2t}{T}\right)$$

$$h(t) = \kappa \tilde{g}(-t) = \kappa A_M V_0 \operatorname{tri}\left(\frac{2t}{T}\right) \rightarrow \text{e simmetrico} \quad h(-t) = h(t)$$

In questo modo non ho c.s.c.

$$C(f) = G(f) L(f) U(f) \text{ e' a rumore zero/zero}$$

$$V_0' = \tilde{g} * h(t) \Big|_{t=0}$$

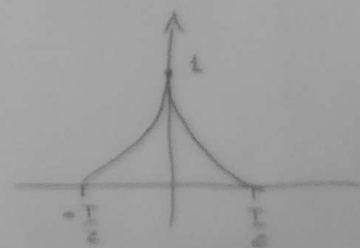
$$\begin{aligned} \tilde{G}(f) H(f) &= A_M V_0 \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2} f\right) \cdot A_M V_0 \frac{T}{2} \operatorname{sinc}^2\left(\frac{T}{2} f\right) \cdot \kappa \\ &= A_M^2 V_0^2 \frac{T^2}{4} \kappa \operatorname{sinc}^4\left(\frac{T}{2} f\right) \end{aligned}$$

$$V_0' = A_M^2 V_0^2 \frac{T^2}{4} \kappa$$

$$\sigma_n^2 = R_0 \int_{-\infty}^{\infty} |h(t)|^2 dt = R_0 \kappa A_M V_0 \int_{-\infty}^{\infty} \operatorname{tri}^2\left(\frac{2t}{T}\right) dt =$$

$$R_0 \kappa A_M V_0 \frac{T}{3}$$

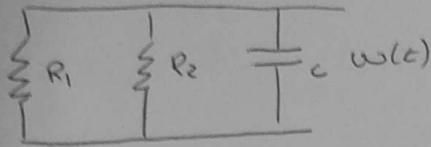
$$P_e = \frac{2M-2}{M} Q\left(\frac{V_0'}{\sigma_n}\right) = \frac{4}{3} Q\left(\frac{V_0'}{\sigma_n}\right)$$



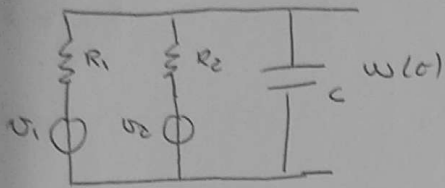
Esercizio 4

$$R_1 \rightarrow T_1 = T_0$$

$$R_2 \rightarrow T_2 = 270$$



↓



Passando alla notazione simbolica:

$$W = V_1 \frac{R_2 // C}{R_2 // C + R_1} + V_2 \frac{R_1 // C}{R_1 // C + R_2}$$

$$R_2 // C = \left(\frac{1}{R_2} + j\omega C \right)^{-1} = \left(\frac{1 + j\omega C R_2}{R_2} \right)^{-1} = \frac{R_2}{1 + j\omega C R_2}$$

$$R_1 // C = \frac{R_1}{1 + j\omega C R_1}$$

$$W_1 = V_1 \frac{\frac{R_2}{1 + j\omega C R_2}}{\frac{R_2}{1 + j\omega C R_2} + R_1} + V_2 \frac{\frac{R_1}{1 + j\omega C R_1}}{\frac{R_1}{1 + j\omega C R_1} + R_2} =$$

$$= V_1 \frac{R_2}{\underbrace{R_2 + j\omega C R_1 R_2 + R_1}_{H_1(f)}} + V_2 \frac{R_1}{\underbrace{R_1 + j\omega C R_1 R_2 + R_2}_{H_2(f)}}$$

$$W W^* = V_1 V_1^* |u_1(f)|^2 + V_2 V_2^* |u_2(f)|^2$$

$$R_W = R_{W1} |u_1(f)|^2 + R_{W2} |u_2(f)|^2$$

$$= 2kT_1 R_1 |u_1(f)|^2 + 2kT_2 R_2 |u_2(f)|^2$$

STABILITÀ BIBO (Bounded Input, Bounded Output)

4

$$\frac{x(nT)}{z(T)} \boxed{h(nT)} \frac{y(nT)}{z(T)}$$

se $|x(nT)| < M$ ($M < +\infty$), vorrei avere $|y(nT)| < M_y$ ($M_y < +\infty$)

Condizione necessaria e sufficiente per la stabilità in senso BIBO è che la risposta impulsiva del filtro sia sommabile:

$$\sum_{k=-\infty}^{+\infty} |h(kT)| < +\infty$$

Dim:

Condizione è sufficiente

$$y(nT) = T \sum_{k=-\infty}^{+\infty} h(kT) x(nT - kT)$$

$$|y(nT)| = \left| T \sum_{k=-\infty}^{+\infty} h(kT) x(nT - kT) \right| \leq T \sum_{k=-\infty}^{+\infty} |h(kT)| |x(nT - kT)| < TM \sum_{k=-\infty}^{+\infty} |h(kT)| < +\infty$$

Condizione è necessaria

L'ingresso $x(nT) = \frac{h^*(-nT)}{|h(-nT)|}$ porta ad un $y(0)$ che diverge

$$|y(0)| = T \sum_{k=-\infty}^{+\infty} \frac{h(kT) \cdot h^*(kT)}{|h(+kT)|} = +\infty \quad \text{se} \quad T \sum_{k=-\infty}^{+\infty} |h(kT)| = +\infty$$

$$\hookrightarrow \frac{|h(kT)|^2}{|h(+kT)|} = |h(kT)|$$

FILTRI CON RISPOSTA IN FREQUENZA RAZIONALE

Si tratta di filtri realizzabili finemente mediante semplici algoritmi.

Filtri REALI $h(nT) = h^*(nT)$

$$H(f) = T \frac{\sum_{k=0}^r b_k e^{-j2\pi f k T}}{\sum_{k=0}^q a_k e^{-j2\pi f k T}}$$

$$H(f) = T \sum_{k=-\infty}^{+\infty} h(kT) e^{-j2\pi f k T}$$

$b_k, a_k \rightarrow$ costanti reali
 $a_0 = 1 \rightarrow$ per definire
 $q \rightarrow$ poli
 $r \rightarrow$ zeri

TRASFORMATA Z

$$H_z(z) = T \sum_{k=-\infty}^{+\infty} h(kT) z^{-k} \quad z \in \mathbb{C}$$

$$H(z) = T \frac{\sum_{k=0}^r b_k z^{-k}}{\sum_{k=0}^q a_k z^{-k}}$$

La trasformata di Fourier equivale alla trasformata Z calcolata per
 $z = e^{j2\pi f T} : H(f) = H_z(z) \Big|_{z=e^{j2\pi f T}}$

Esempio: $h(nT) = p^n \delta(nT) \quad |p| < 1 \quad p \in \mathbb{C}$

$$H(f) = T \sum_{k=0}^{+\infty} p^k e^{-j2\pi f k T} = \frac{T}{1 - p e^{-j2\pi f T}}$$

$$H(z) = \frac{T}{1 - p z^{-1}} = \frac{z T}{z - p} \rightarrow p \text{ è un polo della trasf. zeta}$$

$z=0$ è uno zero della " "

$$H(z) = \frac{T \sum_{k=0}^r b_k z^{-k}}{\sum_{k=0}^q a_k z^{-k}} \quad \underline{a_0 = 1}$$

Essendo $r < q \Rightarrow$ tutti i poli sono distinti

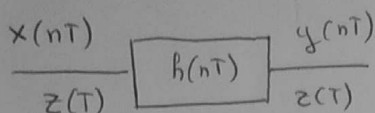
$$H(z) = \frac{T \sum_{k=0}^r b_k z^{-k}}{\prod_{k=1}^q (1 - p_k z^{-1})} = T \sum_{k=1}^q \frac{A_k}{1 - p_k z^{-1}}$$

Se $p_k < 1 \quad \forall k$, la trasformata Z corrisponde alla trasformata zeta di una combinazione lineare di sequenze esponenziali

$$h(nT) = \sum_{k=1}^q A_k p_k^n \delta(nT)$$

FILTRI NUMERICI

Siamo nel dominio discreto del tempo



RISPOSTA IMPULSIVA

Se l'ingresso $x(nT) = \delta_{z(T)}(nT)$ ovvero come uscita $y(nT) = h(nT)$!
 perché essendo l'uscita, la convoluzione tra $x(nT)$ e $h(nT)$, $\delta_{z(T)}$ è

l'elemento neutro della convoluzione quindi $y(nT) = \delta_{z(T)}^{(nT)} * h(nT) = h(nT)$

$$y(nT) = x * h(nT) = T \sum_{k=-\infty}^{+\infty} x(kT) h(nT - kT)$$

STABILITÀ BIBO

Stabilità di un sistema dinamico che per valori limitati dell'ingresso da sempre e solo valori limitati in uscita

Se $|x(nT)| < M \quad \forall n \quad (M < +\infty)$

allora $|y(nT)| < M_y \quad \forall n \quad (M_y < +\infty)$

• Condizione necessaria e sufficiente per avere stabilità BIBO è che la risposta impulsiva del filtro sia sommabile

$$\sum_{k=-\infty}^{+\infty} |h(kT)| < +\infty$$

Dimostrazione

- la condizione è sufficiente

$$y(nT) = x * h(nT) = T \sum_{k=-\infty}^{+\infty} h(kT) x(nT - kT)$$

$$|y(nT)| = \left| T \sum_{k=-\infty}^{+\infty} h(kT) x(nT - kT) \right| \leq T \sum_{k=-\infty}^{+\infty} |h(kT)| |x(nT - kT)| <$$

$$< M T \sum_{k=-\infty}^{+\infty} |h(kT)|$$

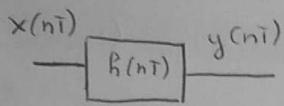
- la condizione è necessaria

se $x(nT) = \frac{h^*(-nT)}{|h(-nT)|}$

$$|y(0)| = T \sum_{k=-\infty}^{+\infty} \frac{h(kT) \cdot h^*(kT)}{|h(kT)|} = +\infty \quad \text{per} \quad T \sum_{k=-\infty}^{+\infty} |h(kT)| = +\infty$$

$$= T \sum_{k=-\infty}^{+\infty} \frac{|h(kT)|^2}{|h(kT)|}$$

divergerebbe



$$Y(f) = H(f) X(f) = X(f) T \frac{\sum_{k=0}^r b_k e^{-j2\pi f k T}}{1 + \sum_{k=1}^q a_k e^{-j2\pi f k T}}$$

$$Y(f) \left(1 + \sum_{k=1}^q a_k e^{-j2\pi f k T} \right) = X(f) T \sum_{k=0}^r b_k e^{-j2\pi f k T}$$

$$Y(f) = -Y(f) \sum_{k=1}^q a_k e^{-j2\pi f k T} + X(f) T \sum_{k=0}^r b_k e^{-j2\pi f k T}$$

$$\downarrow \mathcal{F}^{-1}$$

$$y(nT) = - \sum_{k=1}^q a_k y(nT - kT) + T \sum_{k=0}^r b_k x(nT - kT)$$

È quindi possibile scrivere l'uscita corrente, come combinazione lineare di valori precedenti dell'uscita e dell'ingresso (precedente e corrente)

EQUAZIONE ALLE DIFFERENZE